

MOUNT FOREST, ON, Canada

WMO: 716310

Lat: 43.9895N

Lon: 80.7447W

Elev: 1360

StdP: 13.99

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.2	(c) -0.9	(d) -13.0	(e) 2.9	(f) -5.1	(g) -6.9	(h) 4.0	(i) -0.1	(j) 21.0	(k) 20.8	(l) 19.2	(m) 20.7	(n) 5.3	(o) 340	(p) 0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.5	(c) 83.9	(d) 70.2	(e) 81.2	(f) 68.7	(g) 78.6	(h) 67.2	(i) 72.8	(j) 80.2	(k) 71.1	(l) 78.0	(m) 69.6	(n) 75.9	(o) 7.7	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.5	118.3	76.3	68.9	111.7	74.8	67.3	105.7	73.4	37.4	80.1	35.8	78.1	34.5	75.9	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.6	16.4	14.5	-15.0	87.9	6.1	2.3	-19.4	89.6	-23.0	90.9	-26.4	92.2	-30.8	93.9		
			-14.9	76.6	6.5	2.2	-19.6	78.1	-23.4	79.4	-27.0	80.6	-31.7	82.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.8	18.6	19.8	28.3	41.1	53.4	62.8	66.7	65.3	59.0	47.5	35.7	25.4	(6)
	DBStd	19.36	11.66	10.87	11.39	9.48	8.90	7.10	5.83	5.47	7.59	8.60	9.07	9.32	(7)
	HDD50	4226	974	844	678	297	65	3	0	0	12	155	435	763	(8)
	HDD65	8034	1439	1264	1137	717	376	123	50	64	210	546	880	1228	(9)
	CDD50	1946	0	0	6	30	169	386	517	475	281	76	6	0	(10)
	CDD65	281	0	0	0	1	16	56	102	74	29	3	0	0	(11)
	CDH74	1967	0	0	1	7	132	436	733	454	191	13	0	0	(12)
	CDH80	355	0	0	0	0	18	89	146	73	29	0	0	0	(13)
(14) Wind	WSAvg	6.7	8.0	7.8	7.7	7.8	6.6	5.4	5.1	4.9	5.4	6.7	7.6	7.8	(14)
(15) Precipitation	PrecAvg	38.5	3.3	2.6	2.4	3.0	3.4	3.4	3.6	3.1	3.4	3.2	3.5	3.1	(15)
	PrecMax	48.5	5.0	4.6	5.9	4.9	7.0	7.7	5.9	5.8	5.9	6.1	6.5	5.3	(16)
	PrecMin	27.0	2.0	0.9	1.0	1.1	1.3	1.6	1.2	1.3	1.1	0.9	1.2	1.3	(17)
	PrecStd	5.2	0.9	0.9	1.2	1.3	1.3	1.5	1.5	1.2	1.2	1.5	1.2	1.0	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.8	49.5	66.0	74.5	82.4	86.1	87.3	86.1	83.7	75.8	61.7	52.4	(19)
		MCWB	49.6	47.1	58.8	60.1	66.2	72.1	72.2	70.7	70.1	64.2	54.6	50.5	(20)
	2%	DB	43.0	42.8	56.2	67.2	78.0	82.6	84.0	81.7	79.1	70.0	56.8	45.2	(21)
		MCWB	42.4	40.7	49.8	55.7	64.8	69.2	70.7	69.3	67.7	61.7	52.6	43.6	(22)
	5%	DB	37.9	38.0	49.6	61.7	73.1	79.3	80.9	78.8	75.5	65.4	52.9	40.8	(23)
		MCWB	36.9	35.6	44.9	52.5	61.2	67.2	68.8	68.0	65.6	59.3	49.3	39.4	(24)
	10%	DB	34.0	34.1	44.0	56.9	68.8	75.6	78.0	76.1	71.8	61.0	49.1	36.6	(25)
		MCWB	33.0	32.4	39.9	48.8	58.3	65.1	67.0	66.3	63.6	56.0	46.3	35.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.3	47.6	59.8	62.6	69.8	73.8	76.0	74.8	72.6	67.8	57.4	50.8	(27)
		MCDB	49.7	48.8	66.0	70.1	79.0	82.7	83.7	80.0	80.2	72.8	59.3	51.9	(28)
	2%	WB	42.6	41.0	51.2	57.9	66.8	71.6	73.2	71.8	70.0	63.5	53.4	44.1	(29)
		MCDB	43.1	43.2	54.4	64.7	75.0	80.1	80.4	78.5	75.8	67.7	56.4	45.1	(30)
	5%	WB	37.1	35.8	45.2	54.1	63.9	69.5	71.2	70.0	67.5	59.7	49.6	39.4	(31)
		MCDB	37.7	37.7	49.2	59.7	71.0	76.5	77.7	76.0	72.8	64.7	52.0	40.8	(32)
	10%	WB	33.1	32.6	40.1	49.7	60.5	67.2	69.5	68.3	65.3	56.5	46.1	35.4	(33)
		MCDB	33.8	33.9	43.5	56.3	66.7	73.4	75.6	73.7	70.4	60.4	49.0	36.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.8	13.8	14.8	17.3	18.7	18.1	18.5	18.0	17.9	14.7	11.7	10.4	(35)
		MCDBR	13.6	15.7	20.1	24.8	22.3	21.6	20.7	20.3	21.1	20.2	17.7	13.4	(36)
		MCWBR	13.3	14.1	15.2	16.6	12.6	11.8	10.7	10.6	11.6	12.9	14.3	12.7	(37)
	5% WB	MCDBR	13.7	14.8	18.8	21.9	19.8	18.4	17.3	16.8	17.4	17.9	16.1	13.2	(38)
		MCWBR	13.7	14.2	15.5	16.9	12.7	11.5	10.5	10.4	11.1	12.9	14.1	13.0	(39)
(40) Clear-Sky Solar Irradiance	taub		0.278	0.311	0.312	0.349	0.375	0.400	0.396	0.388	0.358	0.343	0.303	0.276	(40)
	taud		2.336	2.201	2.392	2.356	2.328	2.267	2.302	2.313	2.399	2.426	2.489	2.401	(41)
	Ebn at Noon		269	276	294	290	284	276	276	274	275	264	260	259	(42)
	Edh at Noon		25	35	33	37	39	42	40	38	32	28	22	22	(43)
(44) All-Sky Solar Radiation	RadAvg		405	696	1107	1426	1719	1895	1895	1633	1289	760	445	289	(44)
	RadStd		54	90	120	144	160	126	113	83	120	78	68	37	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+1.04	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (50 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page